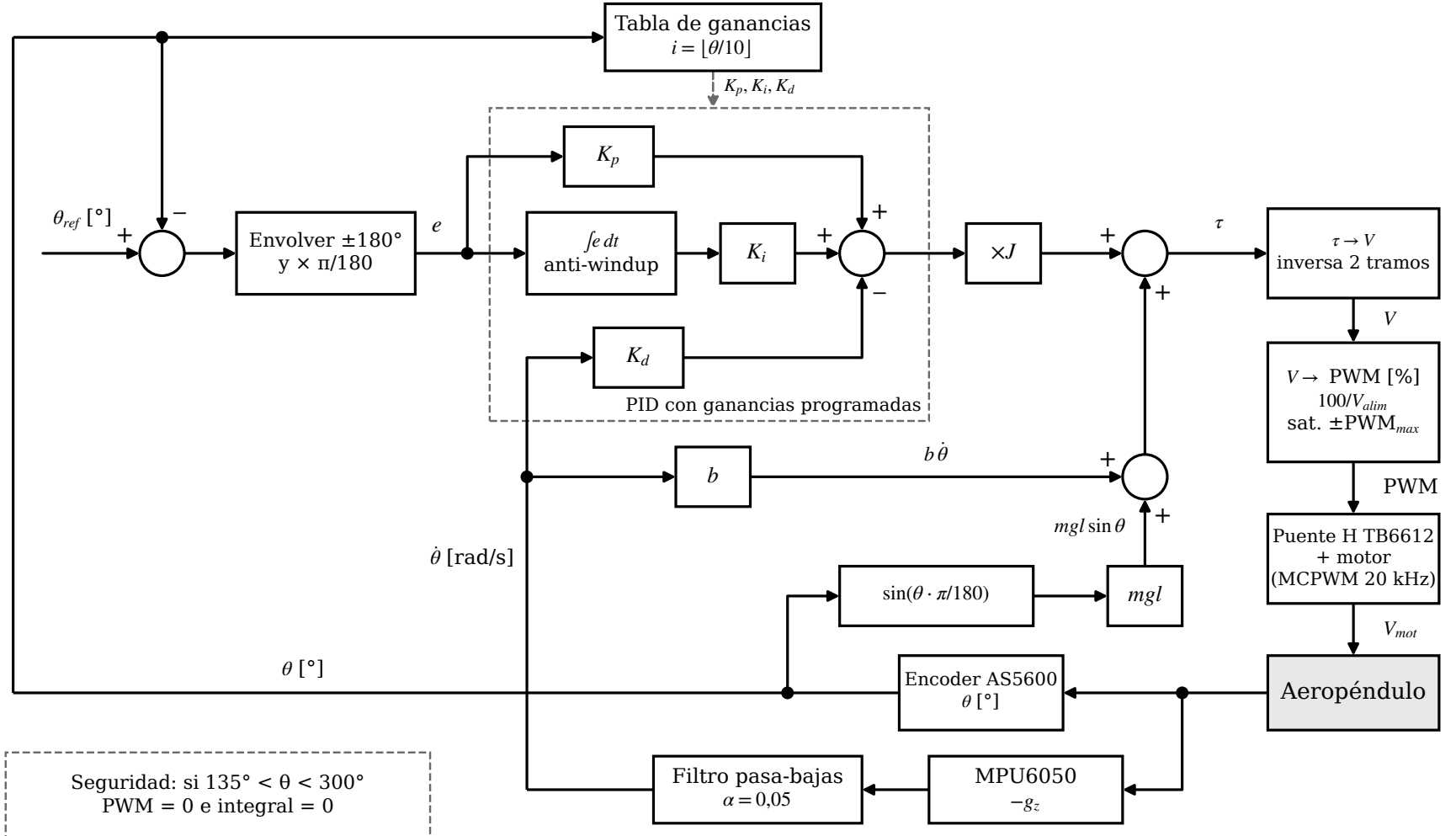
Tabla de ganancias
$i = \lfloor \theta/10 \rfloor$
K_p, K_i, K_d
θ_{ref} [°]
Envolver ±180°
y × π/180
e
K_p
$\int e\,dt$
anti-windup
K_i
K_d
PID con ganancias programadas
$\times J$
τ
$\tau \rightarrow V$
inversa 2 tramos
V
$V \rightarrow$ PWM [%]
$100/V_{alim}$
sat. ±PWM$_{max}$
PWM
Puente H TB6612
+ motor
(MCPWM 20 kHz)
V_{mot}
Aeropéndulo
b
$b\,\dot{\theta}$
$mgl \sin\theta$
$\dot{\theta}$ [rad/s]
$\sin(\theta \cdot \pi/180)$
mgl
θ [°]
Encoder AS5600
θ [°]
Filtro pasa-bajas
$\alpha = 0{,}05$
MPU6050
$-g_z$
Seguridad: si 135° < θ < 300°
PWM = 0 e integral = 0